61. Standby RAM

This MCU provides an on-chip static RAM that can retain data in deep software standby (standby RAM).

61.1 Overview

Table 61.1 lists the specifications of the standby RAM.

Table 61.1 Specifications of Standby RAM

Item	Description
RAM capacity	8 Kbytes
RAM address	000A 4000h to 000A 5FFFh
Access	• Both read and write operations take 3 or 4 cycles of PCLKB when ICLK ≥ PCLKB; 2 or 3 cycles of ICLK are needed when ICLK < PCLKB. • Enabling or disabling of RAM access is selectable.*1 • Endian conforms to the endian setting of the chip. • Non-aligned access is prohibited. If non-aligned access is attempted, correct operation is not guaranteed.
Data retention function	Data can be retained in deep software standby mode
Low-power consumption function	The module-stop state is selectable.

Note 1. Selectable by the SYSCR1.SBYRAME bit. For details on SYSCR1, refer to section 3.2.3, System Control Register 1 (SYSCR1).

61.2 Operation

61.2.1 Data Retention

Whether or not the supply of internal power to the standby RAM continues in deep software standby mode is selectable by the DPSBYCR.DEEPCUT[1:0] bits.

If continuation of the supply of internal power is selected, data in the standby RAM are retained in deep software standby mode.

For details on the DPSBYCR.DEEPCUT[1:0] bits, refer to section 11, Low Power Consumption.

61.2.2 Low Power Consumption Function

Power consumption can be reduced by setting module stop control register C (MSTPCRC) to stop supply of the clock signal to the standby RAM.

If the MSTPCRC.MSTPC7 bit is set to 1, supply of the clock signal to the standby RAM is stopped.

The standby RAM is thus placed in the module-stop state by stopping supply of the clock signals. The standby RAM operates after a reset.

The standby RAM is not accessible if it is in the module-stop state. A transition to the module-stop state should not be made while access to the standby RAM is in progress.

For details on the MSTPCRC register, refer to section 11, Low Power Consumption.